Tutorial 8 - Digital Systems

1. Consider the circuit shown in Figure 1. Identify which FSM (Mealy/Moore) it represents. Also, construct the state table.

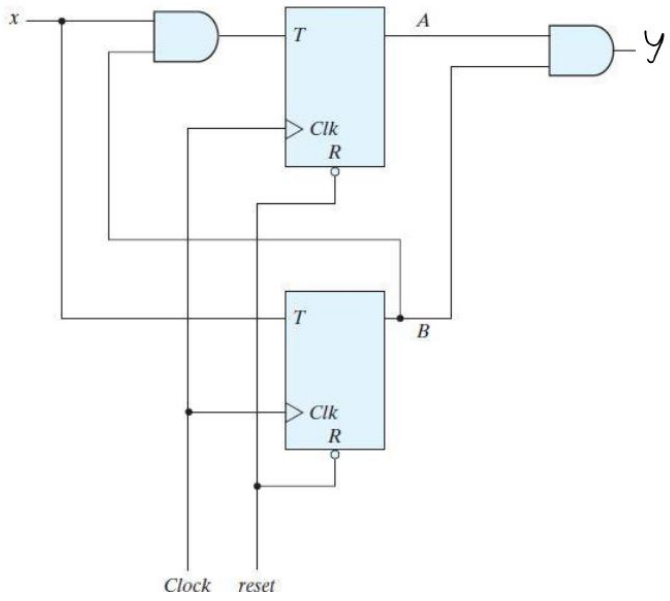


Figure 1: Sequential logic circuit

2. Given below is the state table for a synchronous sequential circuit. Realize the circuit using D flip-flops.

$Q_2Q_1Q_0$	Present State	Next State, Output (for input X = 0)	Next State, Output (for input X = 1)
0 0 0	A	B,0	E,0
0 0 1	B	A,1	C,1
0 1 0	C	B,0	C,1
0 1 1	D	C,0	E,0
1 0 0	E	D,1	A,0

3. Derive the state table and state diagram for the synchronous ciruit shown in Figure 2.

4. A sequence generator is shown in Figure 3. The counter state ($Q_0Q_1Q_2$) is initialized to 010 using the preset/clear inputs. Determine the sequence generated (until it repeats).

5. Consider the state table shown in Figure 4.

   (a) Draw the corresponding state diagram.

   (b) Tabulate the reduced state table.

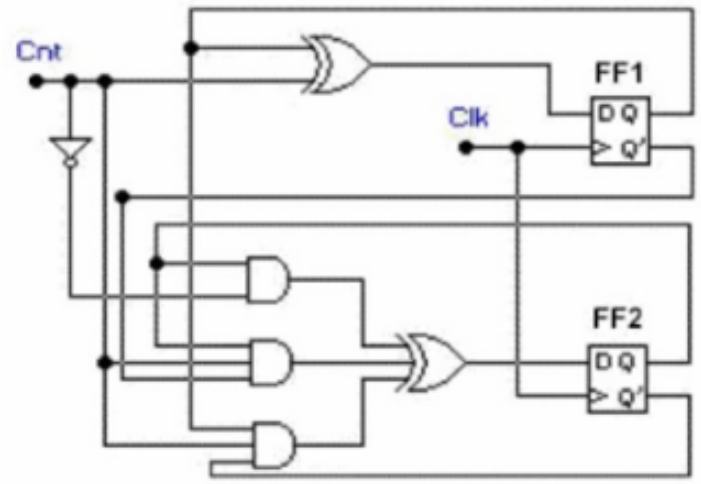


Figure 2: Synchronous circuit

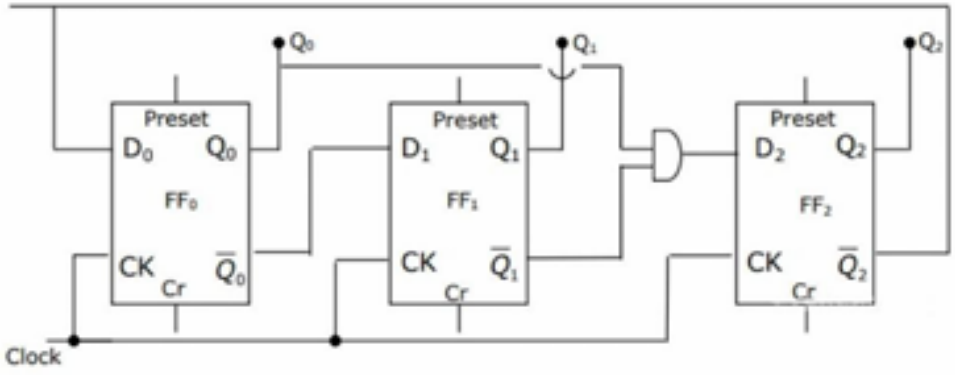


Figure 3: Sequence Generator

(c) Draw the state diagram corresponding to the reduced state table.

6. Realize the state table shown using the minimum number of D flip-flops.

Present State	**Next State**		**Output**	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	e	c	0	1
c	a	d	0	1
d	e	f	0	1
e	a	f	0	0
f	g	d	0	1
g	a	b	0	0

7. For the state diagram shown in Figure 5

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

Figure 4: State table

(a) Write the output sequence for the input sequence: `001010110110110111`

(b) What is the functionality of this FSM (what function does it perform) ?

(c) Identify a flip-flop that can be used to implement the FSM without any external logical gates.

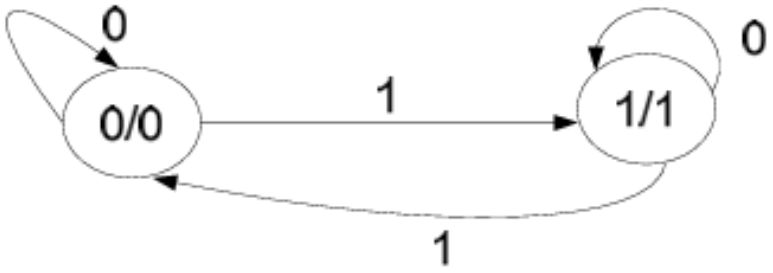


Figure 5: State diagram

8. A combinational circuit has a static 1-hazard if, in response to any single input change and for some combination of propagation delays, the circuit output momentarily goes to 0 when it should remain a constant 1. Consider the circuit shown in Figure 6. Determine if it has a static 1-hazard by drawing a Karnaugh map. Then modify the circuit to remove the hazard.

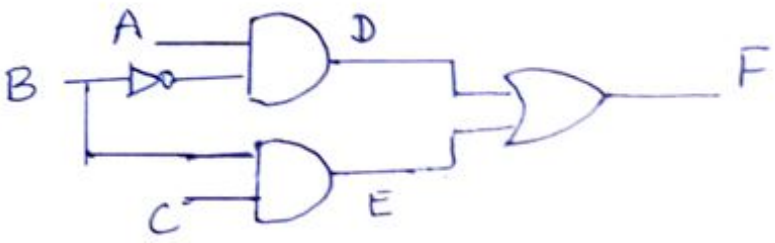
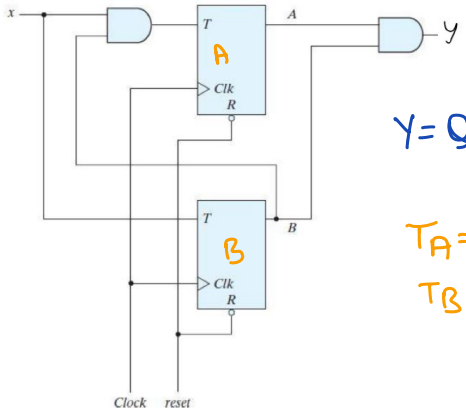


Figure 6: Circuit to check for static 1-hazard

1. Consider the circuit shown in Figure 1. Identify which FSM (Mealy/Moore) it represents. Also, construct the state table.



$Y = Q_A Q_B$

$T_A = x Q_B$

$T_B = x$

It is a moore state machine as the output only depends on present states and not inputs

present		input	next		output
Q_A	Q_B	x	Q_A^+	Q_B^+	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

state table

2. Given below is the state table for a synchronous sequential circuit. Realize the circuit using D flip-flops.

$Q_2Q_1Q_0$	Present State	Next State, Output (for input X = 0)	Next State, Output (for input X = 1)
0 0 0	A	B,0	E,0
0 0 1	B	A,1	C,1
0 1 0	C	B,0	C,1
0 1 1	D	C,0	E,0
1 0 0	E	D,1	A,0

State	Input	Next	Output
000	0	001	0
000	1	100	0
001	0	000	1
001	1	010	1
010	0	001	0
010	1	010	1
011	0	010	0
011	1	100	0
100	0	011	1
100	1	000	0

Q_2^+

Q_2Q_1 / Q_0x	00	01	11	10
00			X	
01	1		X	
11		1	X	X
10			X	X

$$Q_2^+ = \bar{Q}_2\bar{Q}_1\bar{Q}_0x + \bar{Q}_2Q_1Q_0x$$
$$= \bar{Q}_2x[Q_1 \odot Q_0]$$

Q_1^+

Q_0x \ Q_2Q_1	00	01	11	10
00			X	1
01		1	X	
11	1		X	X
10		1	X	X

$$Q_1^+ = Q_1Q_0\bar{x} + Q_1\bar{Q}_0x + \bar{Q}_1Q_0x + Q_2\bar{x}$$

$$= Q_1[Q_0 \oplus x] + x[Q_1 \oplus Q_0] + Q_2\bar{x}$$

Q_0^+

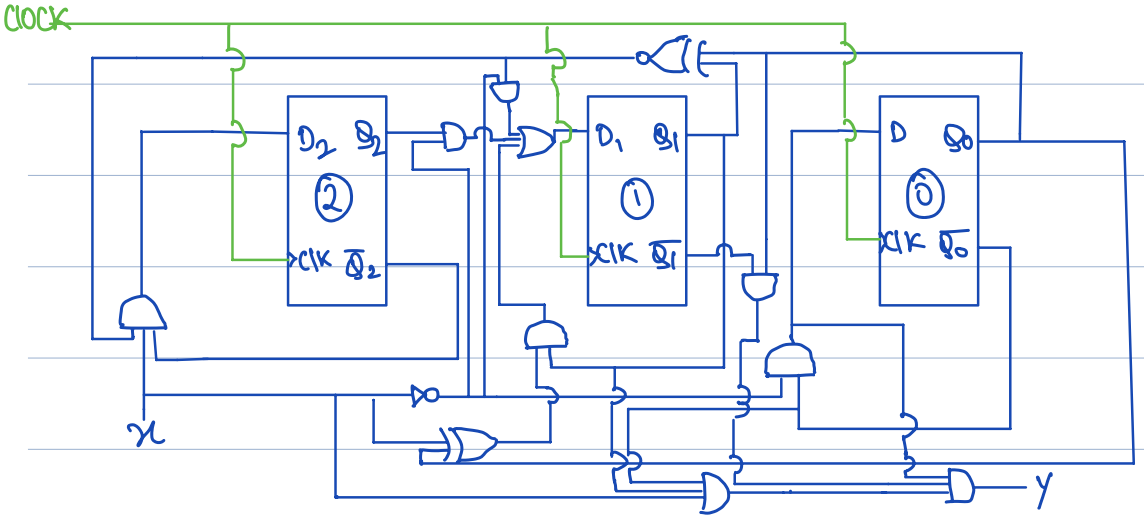
Q_0x \ Q_2Q_1	00	01	11	10
00	1	1	X	1
01			X	
11			X	X
10			X	X

$$Q_0^+ = \bar{Q}_0\bar{x}$$

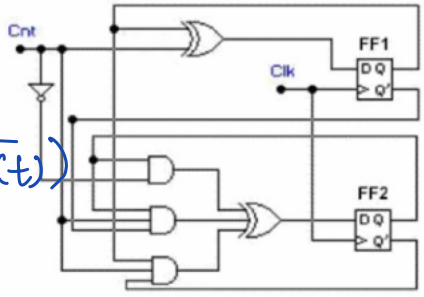
y

Q_0x \ Q_2Q_1	00	01	11	10
00			X	1
01		1	X	
11	1		X	X
10	1		X	X

$$y = Q_2\bar{x} + \bar{Q}_1Q_0 + \bar{Q}_0xQ_1$$



3. Derive the state table and state diagram for the synchronous ciruit shown in Figure 2.



$$Q_1(t+1) = Q_1(t) \oplus C$$

$$Q_2(t+1) = (Q_2(t).\overline{C}) \oplus (Q_2(t) C \overline{Q_1(t)}) \oplus (Q_1(t).\overline{Q_2(t)} C)$$

$$= (\overline{Q_2(t)} + C).(Q_2(t) C \overline{Q_1(t)}).(\overline{Q_1(t)} + Q_2(t) + \overline{C})$$

$$+ (Q_2(t).\overline{C}).(\overline{Q_2(t)} + \overline{C} + Q_1(t)).(\overline{Q_1(t)} + Q_2(t) + \overline{C})$$

$$+ (\overline{Q_2(t)} + C).(\overline{Q_2(t)} + \overline{C} + Q_1(t))(Q_1(t)\overline{Q_2(t)} C)$$

$$+ (Q_2(t)\overline{C})(Q_2(t).C.Q_1(t))(Q_1(t).\overline{Q_2(t)}.C)$$

$$= Q_2(t).C.\overline{Q_1(t)} + \overline{C}\,\overline{Q_1(t)}\,Q_2(t) + \overline{C}\,Q_2(t)$$

$$+ Q_1(t)\,Q_2(t)\,\overline{C} + Q_1(t)\,\overline{Q_2(t)}\,C$$

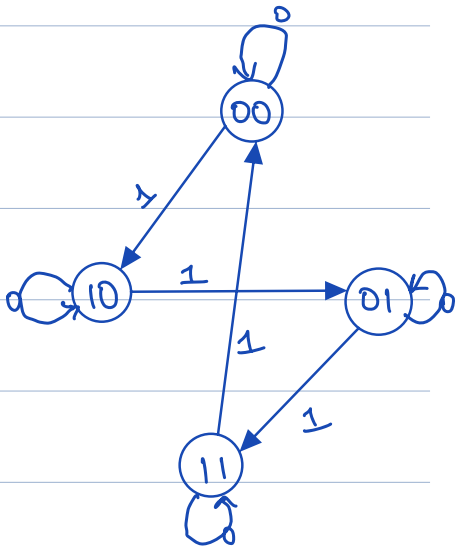
$$= \overline{C}\,Q_2(t) + \overline{Q_1(t)}\,Q_2(t) + Q_1(t)\,\overline{Q_2(t)}\,C$$

$$= Q_2(t)[\overline{C} + \overline{Q_1(t)}] + \overline{Q_2(t)}[C.Q_1(t)]$$

$$= Q_2(t)[\overline{C.Q_1(t)}] + \overline{Q_2(t)}[C.Q_1(t)]$$

$$= (C Q_1(t)) \oplus Q_2(t)$$

present		Input	Next	
Q_1	Q_2	C	Q_1^+	Q_2^+
0	0	0	0	0
		1	1	0
0	1	0	0	1
		1	1	1
1	0	0	1	0
		1	0	1
1	1	0	1	1
		1	0	0

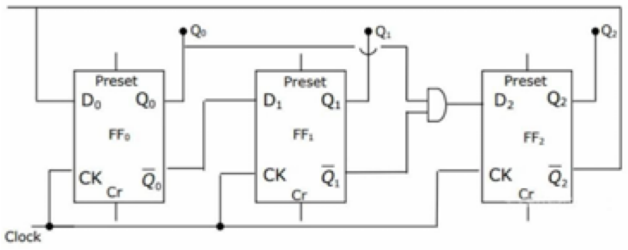


4. A sequence generator is shown in Figure 3. The counter state ($Q_0Q_1Q_2$) is initialized to 010 using the preset/clear inputs. Determine the sequence generated (until it repeats).

$D_0 = \overline{Q_2} = Q_0^+$

$D_1 = \overline{Q_0} = Q_1^+$

$D_2 = Q_0\overline{Q_1} = Q_2^+$



	Q_0	Q_1	Q_2	Q_0^+	Q_1^+	Q_2^+
n=1	0	1	0	1	1	0
n=2	1	1	0	1	0	0
n=3	1	0	0	1	0	1
n=4	1	0	1	0	0	1
n=5	0	0	1	0	1	0

5. Consider the state table shown in Figure 4.

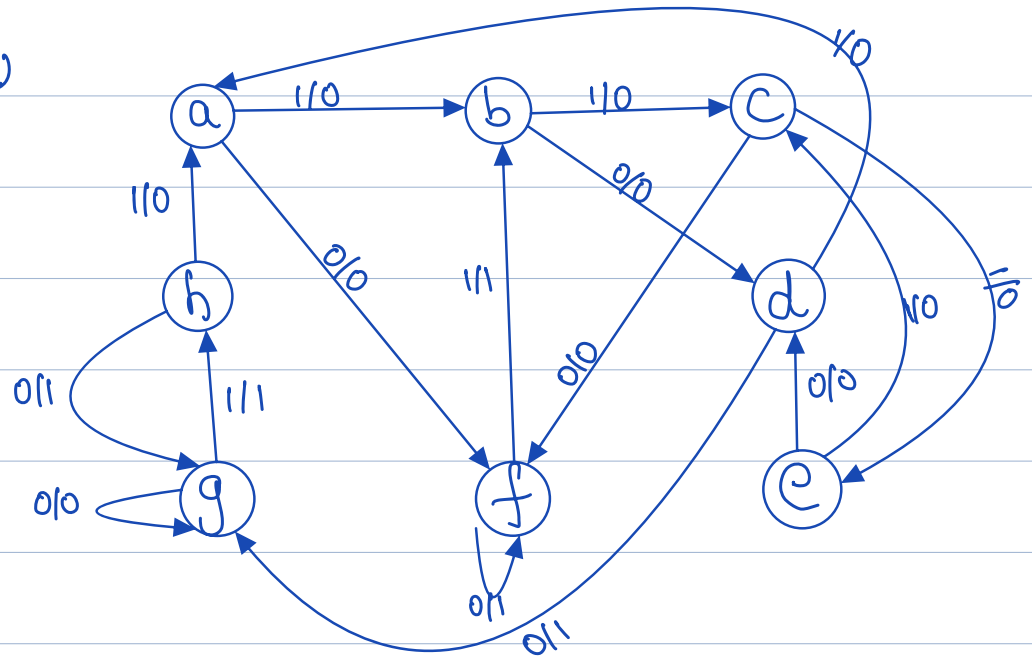
(a) Draw the corresponding state diagram.

(b) Tabulate the reduced state table.

(c) Draw the state diagram corresponding to the reduced state table.

	Next State		Output	
Present State	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

(a)



(b) d and h are equivalent, so are b and e

⇒

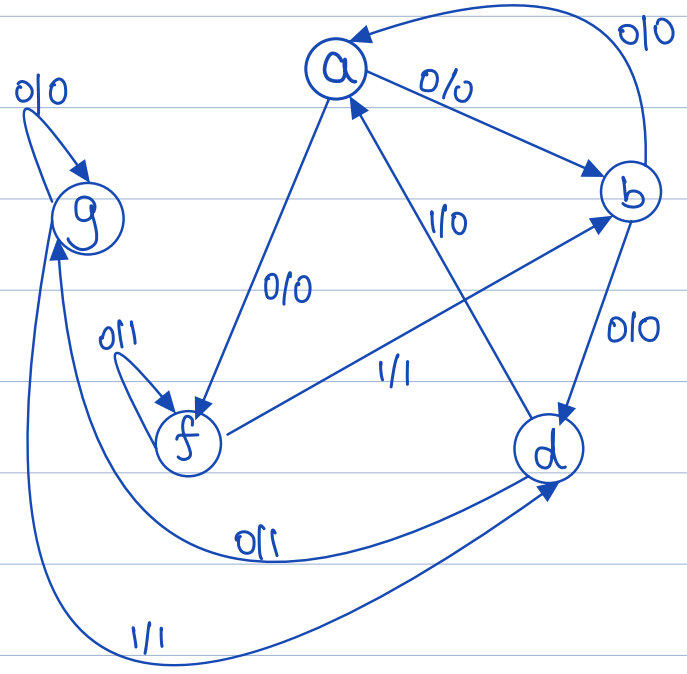
present	next $x=0$	next $x=1$	output $x=0$	output $x=1$
a	f	b	0	0
b	d	c	0	0
c	f	b	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

Now we can see that a and c are equivalent

⇒

present	next $x=0$	next $x=1$	output $x=0$	output $x=1$
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

(c)



6. Realize the state table shown using the minimum number of D flip-flops.

Present State	Next State		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	a	b	0	0
b	e	c	0	1
c	a	d	0	1
d	e	f	0	1
e	a	f	0	0
f	g	d	0	1
g	a	b	0	0

LSB: Q_0
MSB: Q_2

g and a are equivalent

present	Next x=0	Next x=1	Output x=0	Output x=1
a	a	b	0	0
b	e	c	0	1
c	a	d	0	1
d	e	f	0	1
e	a	f	0	0
f	a	d	0	1

now c and f are equivalent

present	Next x=0	Next x=1	Output x=0	Output x=1
a	a	b	0	0
b	e	c	0	1
c	a	d	0	1
d	e	c	0	1
e	a	c	0	0

Now b and d are equivalent

present	Next $x=0$	Next $x=1$	Output $x=0$	Output $x=1$
a	a	b	0	0
b	e	c	0	1
c	a	b	0	1
e	a	c	0	0

$Q_1 \rightarrow MSB$

$a = 00 \quad b = 01 \quad c = 10 \quad e = 11$

Q_1	Q_0	x	Q_1^+	Q_0^+	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	0	0	0
1	1	1	1	0	0

Q_1^+:

x \ Q_1Q_0	00	01	11	10
0				1
1			1	1

$Q_1^+ = Q_1\bar{Q_0} + xQ_1$

Q_0^+:

x \ Q_1Q_0	00	01	11	10
0		1		
1	1			1

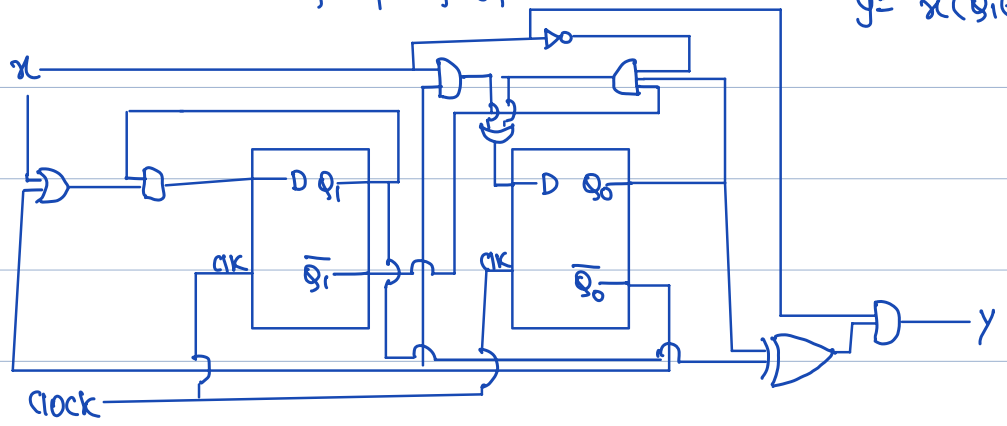
$Q_0^+ = x\bar{Q_0} + \bar{x}\bar{Q_1}Q_0$

y:

x \ Q_1Q_0	00	01	11	10
0				
1		1		1

$y = x(Q_1 \oplus Q_0)$

We need 2 flip flops

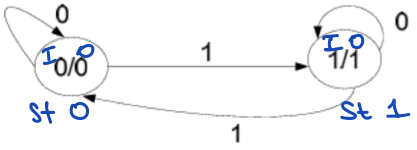


7. For the state diagram shown in Figure 5

(a) Write the output sequence for the input sequence: 001010110110110111

(b) What is the functionality of this FSM (what function does it perform) ?

(c) Identify a flip-flop that can be used to implement the FSM without any external logical gates.



when the input is 0 at State 0, it stays at state 0, with output as 0

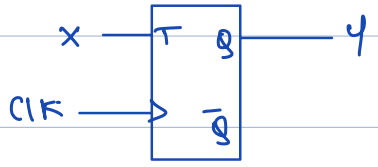
when the input is 1 at State 0, it goes to state 1 with output 0

when the input is 0 at State 1, it goes to state 0 with output 1

when the input is 1 at State 1, it stays at state 1, with output as 1

(b) detects odd number of 1's of the sequence (odd-one-detector)

(c) Since it toggles between State 0 and 1 when input is 1 and remains at the same state when input is 0, best match is a T-flipflop



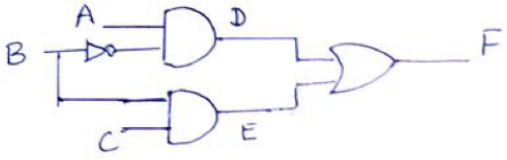
(a)

Input bit	Initial State	Final State	Output bit
0	0	0	0
0	0	0	0
1	0	1	0
0	1	1	1
1	1	0	1
0	0	0	0
1	0	1	0
1	1	0	1
0	0	0	0
1	0	1	0
1	1	0	1
0	0	0	0
1	0	1	0
1	1	0	1
1	0	1	0

8. A combinational circuit has a static 1-hazard if, in response to any single input change and for some combination of propagation delays, the circuit output momentarily goes to 0 when it should remain a constant 1. Consider the circuit shown in Figure 6. Determine if it has a static 1-hazard by drawing a Karnaugh map. Then modify the circuit to remove the hazard.

$D = A\bar{B}$ $F = A\bar{B} + BC$

$E = BC$



A \ BC	00	01	11	10
0			1	
1	1	1	1	

$(A = C = 1)$

If we change B from 0 to 1; $A\bar{B}$ goes to 0 BC goes to 1. But due to delay in gates the output will be like 1, 0 (momentarily) then 1 again. To fix this we can add a redundant term i.e their overlap AC

$F = A\bar{B} + BC + AC$

This will remove the glitch

